

Uncertainty Aware Inference

Evaluating how PTQ affects LLM confidence calibration and developing uncertainty-based routing for inference gains

MENTOR PROJECT · TEAM C (17)

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ADVISOR & DATE

Advised by Dr. Ruchi Mahindru (IBM Research)
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LINKS

github.com/chuanbinp/uncertainty-aware-inference · wandb.ai/Uncertainty_Aware_Inference_Lab/UAI_Project

Why this matters.

WHY THIS MATTERS

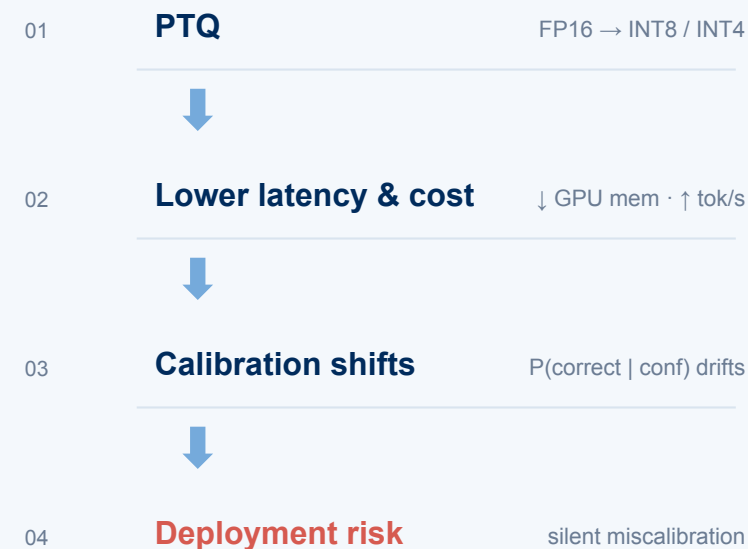
- LLM inference is expensive; **PTQ** cuts GPU cost, latency, and energy.
- Production systems increasingly **rely on model confidence**, not accuracy alone.
- Quantization may preserve accuracy while distorting probability estimates, so its **confidence no longer reflects when it is actually correct**
- Especially **risky in high-stakes domains** like biomedical QA, finance, and autonomous decision systems.

PROBLEM STATEMENT

- **Gap:** Standard PTQ evaluation emphasizes throughput and accuracy, but often overlooks calibration quality; prior work¹ shows quantization can shift predictive confidence
- **Question:** How do GPTQ, AWQ, and NF4 affect confidence calibration for Llama-2 13B?
- **Goal:** Characterize the tradeoff among throughput, accuracy, and calibration quality, and test when uncertainty-aware routing remains favorable

Reference: Proskurina et al., "When Quantization Affects Confidence of Large Language Models?" NAACL, 2024.

DEPLOYMENT RISK IN ONE DIAGRAM



Experimental Setup

MODEL

Llama-2 13B

40 layers · 13B parameters

Config	Bits
FP16	16
GPTQ-INT8	8
GPTQ-INT4	4
AWQ-INT4	4
NF4	4

Implemented through:

- HF
- bitsandbytes
- AutoGPTQ
- AutoAWQ

DATASET

3 QA benchmarks

Full datasets · seed = 42

HellaSwag

Commonsense MCQ · n=10,042

TriviaQA

Factual QA · n=17,944

PubMedQA

Biomedical OOD · n=1,000

Designed to stress different uncertainty regimes.

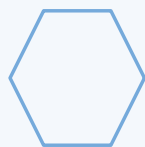
HARDWARE

NVIDIA A100, 40 GB

Single-GPU generation runs

Batch size	1
Inference	HF generate()
Determinism	seed = 42

Evaluation Methodology



01 — PTQ SWEEP

Post-training quantization sweep

Evaluate FP16, INT8, and INT4 quantization (GPTQ, AWQ, NF4) to reduce inference cost while preserving accuracy.

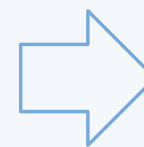


02 — CALIBRATION

Calibration-aware evaluation + Pareto Frontiers

Measure ECE to capture when model confidence diverges from true correctness.

Compute Pareto frontiers across inference cost, task accuracy, and calibration error



03 — ROUTING

Uncertainty-based routing

Route high-confidence queries to cheap models and escalate uncertain ones to FP16.

Experiment Tracking + Key Metrics

TOOLS

HF logs	HuggingFace inference logs from generate()
eval_utils	Custom evaluation pipeline (eval_utils.py)
artifacts	Saved JSON + .pt tensors per run
tracking	Weights & Biases dashboards for every metric

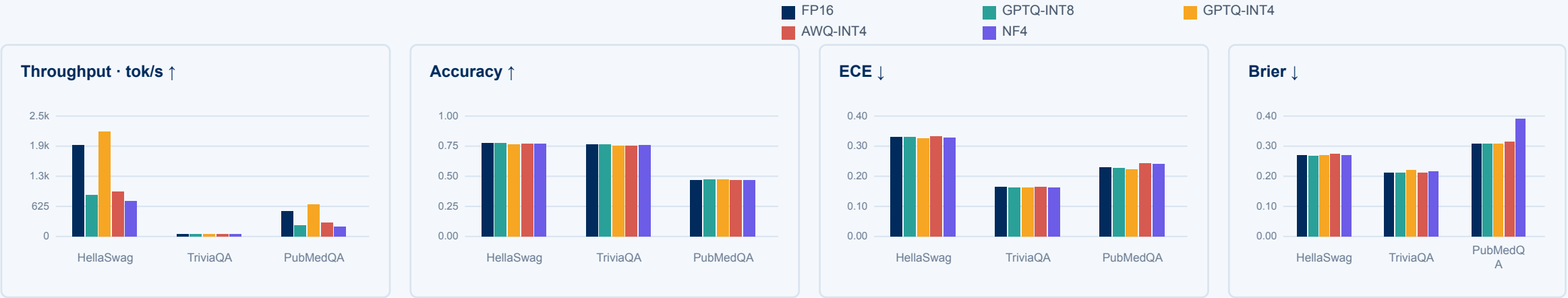
METRICS CAPTURED

throughput	Tokens / second
accuracy	Task accuracy, per dataset
calibration	ECE · MCE · Brier Score
distributions	per token entropy

Comparing metrics across quantization methods and three datasets.

Five Llama-2 13B configurations × three benchmarks. We track throughput, accuracy, ECE, and Brier — and where they diverge.

Comparing metrics between quant methods across 3 datasets.



Config	HellaSwag tok/s	HellaSwag Acc	HellaSwag ECE	HellaSwag Brier	TriviaQA tok/s	TriviaQA Acc	TriviaQA ECE	TriviaQA Brier	PubMedQA tok/s	PubMedQA Acc	PubMedQA ECE	PubMedQA Brier
FP16	1903	0.773	0.331	0.269	39.29	0.767	0.164	0.211	528	0.470	0.229	0.307
GPTQ-INT8	865	0.773	0.331	0.268	18.80	0.767	0.162	0.211	234	0.473	0.228	0.308
GPTQ-INT4	2180	0.766	0.327	0.269	41.74	0.754	0.162	0.220	661	0.472	0.222	0.308
AWQ-INT4	928	0.772	0.332	0.274	22.82	0.752	0.165	0.211	289	0.470	0.242	0.314
NF4	735	0.769	0.329	0.270	19.91	0.757	0.162	0.216	201	0.467	0.240	0.390

Results

Live demo.

Where Llama-2 13B sits on the Pareto frontier.

SETUP

- 5 quantization configs across 3 datasets.
- Metrics: accuracy, ECE, Brier, throughput.
- Cross-model Pareto analysis: 15 configs.
- Routing simulation: threshold sweep across 3 strategies.

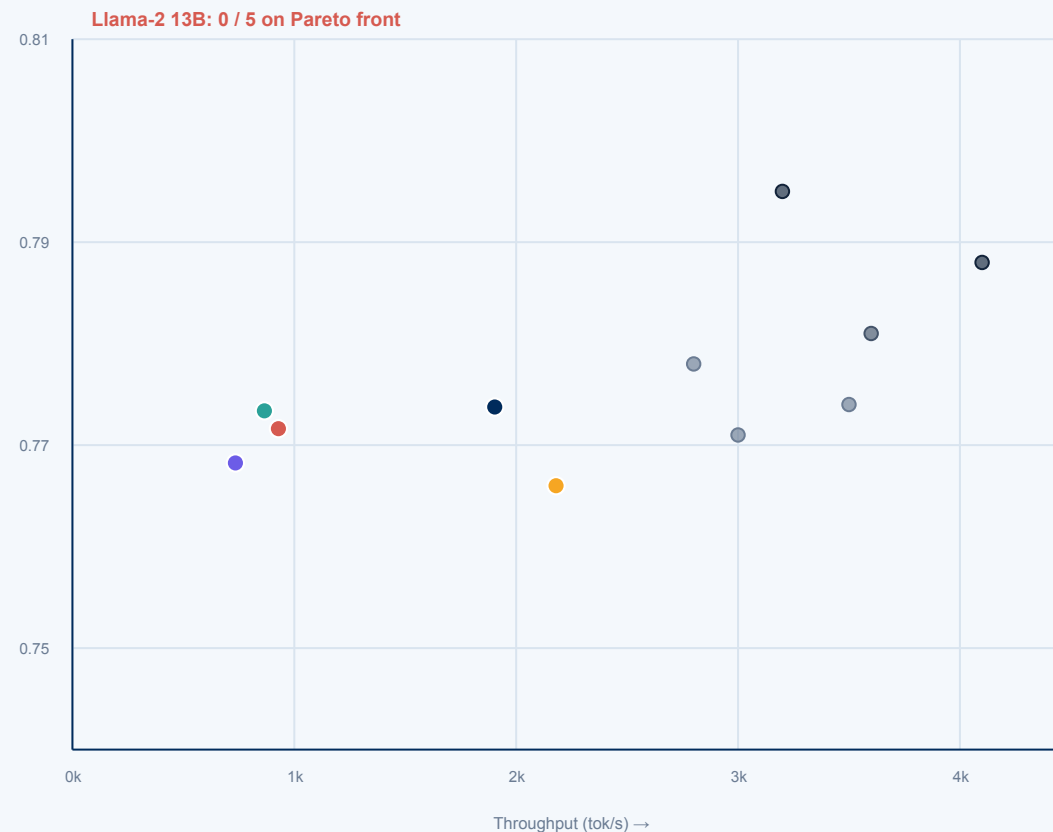
FINDING · PARETO

Llama-2 13B has zero Pareto-optimal configs.

FINDING · ROUTING

Cost savings capped at ~20% due to throughput constraints.

Pareto frontier · accuracy vs throughput



What worked, and what did not.

What worked

- ✓ Calibration metrics reveal failures accuracy hides.
- ✓ GPTQ-INT4 stays Pareto-competitive.
- ✓ Pareto frontiers expose deployment winners.

What did not

- ✗ AWQ was not architecture-robust.
- ✗ Larger models do not imply better calibration.
- ✗ Routing conclusions remain preliminary.

Four directions for follow-up work.

01 **Re-run routing simulation**

02 **Use semantic entropy as the routing signal**

03 **Optimize quantization directly for calibration**

04 **Finalize entropy analysis for future inclusion in report**

Contributions

Member	Primary contributions	Tools / artifacts
Sanket Makkar	Running Int4, int8 experiments	TeamC/full_results/llama2-13b-gptq-int4 TeamC/full_results/llama2-13b-gptq-int8
Chuan Bin Phoe	Experiment and Pareto Dashboarding scripts, Report	TeamC/configs.py, TeamC/run_eval.py, TeamC/llama2_13b.ipynb TeamC/pareto_script.py, TeamC/results/pareto
Samuel Lee	Experiment, Report, Pareto	TeamC/run_eval.py, TeamC/llama2_13b.ipynb, TeamC/full_results/llama2-13b-awq-int4 TeamC/full_results/llama2-13b-nf4, TeamC/full_results/llama2-13b_16bit_fp16
Yechan Jeon	Pareto, Routing Simulation, Report, Slide	TeamC scripts implementations

Collaboration: weekly meetings via Zoom, shared Whatsapp channel + email chains, code review on every PR.

Q & A

Thank you.

Questions?